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Financial distress among cancer survivors in Appalachian Kentucky

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Abstract

Background: Rural residence may exacerbate cancer-related financial distress. Limited research has focused on Appalachian cancer survivors' experience with financial distress.

Aims: The primary aim of this study was to estimate the prevalence of financial distress among cancer survivors residing in Appalachian Kentucky with a specific focus on the impact of rurality and to elucidate the risk factors impacting financial distress among this population.

Methods and Results: Appalachian Kentucky residents were sampled for receipt of a health survey based on county-level rurality. Analyses describe the prevalence and predictors of financial distress among cancer survivors. Subsequent analyses were conducted with Health Information National Trends Survey (HINTS) data to compare local versus national estimates of financial distress. Almost two-thirds of Appalachian survivors reported financial distress compared to one-third of the HINTS sample. Appalachian survivors residing in the most rural counties reported higher distress; this finding was not supported in the national sample. In multivariable analyses, gender, current age, and household income were associated with financial distress among Appalachians; only income was significant among the national sample.

Conclusion: Appalachian cancer survivors have higher than national estimates of financial distress; rurality and socioeconomics are drivers of this disparity.

KEYWORDS

Appalachian region, financial distress, neoplasms, rural health, surveys and questionnaires, survivorship

1 | INTRODUCTION

A cancer diagnosis not only impacts an individual's physical and mental health, but may also significantly impact their financial well-being.[1,2] Studies have found that cancer survivors experience acute and/or chronic financial distress following their diagnosis, subsequent treatment, long-term management of side effects, and continued follow-up care.[3-9] Specifically, survivors have a higher prevalence of: out-of-pocket medical costs, out-of-pocket costs as a percentage of income, employment loss, lost wages and reduced benefits, asset depletion and medical debt, difficulty paying for non-medical bills, bankruptcy, financial worry, and overall financial hardship.[1,2,7-9] In turn, these material conditions and psychological responses – coupled with the cancer itself – may negatively impact quality of life, treatment choices, medication adherence, symptom burden, and overall cancer survival.[1,10-13]

Understanding financial distress among special populations of cancer survivors is important especially given the disparities of those

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residing in rural areas of the U.S.[14] Compared to urban locales, rural communities commonly experience higher rates of cancer incidence and mortality, disparate socioeconomic and education outcomes, health insurance challenges, geographic isolation, transportation barriers, and difficulties accessing oncology and social services, among other issues,[15-17] which may further exacerbate the financial hardship of being diagnosed with cancer.[5,18-20] For example, McDougall et al. found that rural colorectal cancer survivors in New Mexico were twice as likely as urban survivors to report treatment-related financial hardship.[5] Research also suggests financial burden may negatively impact rural cancer patients' ability to obtain medical care, afford medications, participate in long-term cancer surveillance (e.g., colonoscopy), and afford short- and long-term out-of-pocket medical costs.[5,18,21] Financial hardship has also been reported among Latino cancer patients living along the rural U.S.-Mexico border in southern California[22] as well as older African American cancer survivors residing in rural Virginia.[23]

Of particular interest, limited research has explored perceptions of financial hardship among rural cancer patients residing in Appalachia, a unique 13-state geo-politically defined region of the U.S. noted for significant cancer and socioeconomic disparities.[24,25] Indeed, some of the nation's highest cancer incidence and mortality rates are found in the 54 eastern Kentucky counties designated as Appalachian.[26] Many of these same Kentucky counties are among the most medically underserved, geographically isolated, disease-burdened, and economically disadvantaged communities in the U.S.[27,28] Notably, 82% of Appalachian Kentucky is characterized as rural; median household income in Appalachian Kentucky is 40% less than the national median; 27% of Appalachian Kentucky households live in poverty; and only 14% of adults age 25 and older in the region have completed a bachelor's degree or higher.[27,28] Appalachian Kentucky also faces shortages in primary care physicians, mental health providers, specialty physicians, and dentists.[28] This perfect storm of rurality, lower socioeconomic and educational outcomes, and limited resources places Appalachian Kentucky residents in jeopardy of significant financial distress following the diagnosis of cancer, yet there is limited understanding of this phenomena across the region.

Therefore, the primary aim of this study was to estimate the prevalence of financial distress among cancer survivors residing in Appalachian Kentucky with a specific focus on the impact of rurality and to elucidate the risk factors impacting financial distress among this population. Results can be used to inform future interventional research, outreach, and programming tailored to the region. In a secondary aim of the study, we compared findings from Appalachian Kentucky to the National Cancer Institute's (NCI) Health Information National Trends Survey (HINTS) to articulate differences in national estimates of rurality and financial distress among cancer survivors as compared to localized estimates.

2 | METHODS

In 2016, NCI awarded 15 administrative supplements to designated cancer centers focused on the implementation of local population health assessments to help define centers' catchment area communities.[29] As a result of this initiative, the University of Kentucky (UK) Markey Cancer Center administered a health-related survey – ASK: Assessing the Health Status of Kentucky – during summer 2017 in the 54 counties of eastern Kentucky designated as Appalachian by the Appalachian Regional Commission. The ASK survey assessed a range of topics including: cancer knowledge and beliefs, cancer screening and other health behaviors, cancer history, and sociodemographics (e.g., age, gender, ethnicity/race, education, annual household income). All study procedures were approved by the UK Institutional Review Board (IRB).

2.1 | Research procedures

ASK survey methodology included use of an address-based sampling database, which generated a stratified random sample of 3,200 Appalachian Kentucky addresses, which were grouped into four geographic strata based on the 2013 U.S. Department of Agriculture's Rural-Urban Continuum Codes (RUCC) (Table 1).[30] Strata I included counties with RUCC codes 1-3, which are considered metro counties, while Strata II (RUCC 4-5), Strata 3 (RUCC 6-7), and Strata 4 (RUCC 8-9) included counties with population sizes of 20,000 or more, 2,500-19,999, and less than 2,500, respectively. To assure adequate sample size for strata-specific analysis, Strata 1 and 2 were oversampled and Strata 3 was under-sampled based on the 2015 U.S. 5-year American Community Survey (ACS) population estimates.[31]

Similar to the publically-available HINTS sampling frame and mailing procedures,[32] the ASK data collection methods included a prenotice postcard; a first-class mailing of a cover letter, $2 incentive, and paper survey; a reminder postcard; and a second, final mailing of a cover letter and survey. The initial cover letter included the UK IRB approval stamp and study identification number (16-0876) along with specific language that: [1] identified the funding source for the research study (NCI), [2] reiterated participation was voluntary and that there were no known risks to participating, [3] described processes for data confidentiality and aggregation of study results, and 4) provided telephone numbers for the principal investigator (RV) and the UK Office of Research Integrity. The Next-Birthday Method, also described in the study cover letters, identified which adult should answer the questionnaire if there was more than one adult in the household. Completion of the survey acted as study consent. Eight hundred and six surveys were returned for an overall response rate of 25% (Table 1).

To assure the quality of the data, double data entry was used and key data items (e.g., number of adults in the household, county of residence, gender, race) were cross-examined to check for discrepancies. Invalid data values were imputed with valid values based on other data items or noted as missing if valid values were not available. A final weight for each response was calculated based on non-response rates, strata weight, number of adults in a household, and the corresponding background population from the 2015 ACS estimates.

TABLE 1 ASK sampling frame and response rates

	Original address sampling frame	Expected survey response (%)	Undeliverable addresses (%)	Actual survey response (%)	Valid survey data
Strata 1 (codes 1-3, metro)	530	160 (30)	55 (10)	135 (25)	134
Strata 2 (codes 4-5, slightly rural)	640	160 (25)	55 (9)	165 (26)	164
Strata 3 (codes 6-7, rural)	1,120	280 (25)	135 (12)	257 (23)	253
Strata 4 (codes 8-9, completely rural)	910	200 (22)	97 (11)	249 (27)	247
Total	3,200	800 (25)	342 (11)	806 (25)	798

The weight was calibrated to the total corresponding ACS background population across the dimensions of gender and age. The final dataset included 798 valid survey responses.

2.2 | ASK and HINTS survey items and response categories

Related to the current study on cancer-related financial hardship, the ASK survey replicated two HINTS questions that assess [1] respondents' history of cancer ("Have you ever been diagnosed as having cancer? Yes, No") and among those with a history of cancer, [2] the impact of cancer on their finances ("Looking back, since the time you were first diagnosed with cancer, how much, if at all, has cancer and its treatment hurt your financial situation? Not at all, A little, Some, A lot"). Of the 798 valid survey responses, 793 included an answer for the cancer history question; 179 individuals indicated a history of cancer and served as the primary analytic sample.

In comparing ASK results to nationally representative findings, we utilized NCI's publically-available HINTS 5, Cycle 1 data, which was collected from civilian non-institutionalized U.S. adults aged ≥18 between January-May 2017.[33] A total of 13,360 sampling addresses, stratified by two concentrations of minority populations (high/low), were selected; a total of 3,347 questionnaires were returned with 3,285 deemed eligible for inclusion in the final dataset. The overall response rate was 32.4%.[34] In total, 504 individuals responding to the HINTS survey indicated a history of cancer and served as the secondary analytic sample.

The majority of survey items examined in this study were available in both ASK and HINTS datasets. The following variables were included in the analysis and in some cases, include collapsed response categories based on sample distributions: geographic strata [1-4], gender (male, female), race (White, non-White), current age (18-49, 50-64, 65+ years), age at cancer diagnosis (18-49, 50-64, 65+ years), years since cancer diagnosis (0-5, 6-10, 10+ years), occupational status (employed, retired, disabled, other), marital status (married/living as married, divorced/separated, single/widowed), educational attainment (high school education or less, some college education, college graduate or more), insurance status (private, Medicare, other public, none), and household income ($0-$19,999, $20,000-$49,999, $50,000-$99,999, $100,000+). We also assessed two questions related to financial distress in general: "Was there a time in the past 12 months when you needed to see a doctor but could not because of cost?" (Yes, No) and "Which one of these comes closest to your own feelings about your household's income these days?" (living comfortably on present income, getting by on present income, finding it difficult on present income, and finding it very difficult on present income).

3 | STATISTICAL ANALYSIS

For both surveys, weighted univariate descriptive statistics were calculated for the overall sample of cancer survivors and weighted bivariate analyses compared differences in select participant characteristics by cancer-related financial distress. We first conducted analysis of the four response categories of financial distress for both survey datasets (Supplementary Tables 1 and 2). Subsequently, we dichotomized financial distress into two categories: "Not at all", indicating no distress, and an aggregation of "a little", "some" or "a lot" responses, which indicated distress. Due to small sample sizes in the HINTS dataset and similarity in characteristics in the ASK data, we collapsed Strata 3 and 4 into one category so appropriate comparisons could be made. To identify how the continuum of rurality impacted financial distress among cancer survivors responding to ASK as compared to HINTS, a bivariate analysis of financial distress by survey source and geographic strata was performed. In addition, weighted logistic regression was conducted while controlling for survey variables described above for ASK and HINTS data separately. A stepwise selection approach was used to identify the best and final models for ASK and HINTS, which included only statistically significant variables in at least one of the models. All statistical tests were two-sided with a significance level of 0.05. All data analyses were performed using SAS 9.4 (SAS Institute, Inc., Cary, NC, USA).

4 | RESULTS

4.1 | ASK survey findings

Based on ASK survey results presented in Table 2, over three-fourths (77%) of the Appalachian survivors resided in counties designated as Strata 3 or 4. Slightly more women than men were represented in

TABLE 2 ASK: correlation between cancer-related financial distress and participant characteristics

	Total		No financial distress		Financial distress		
	N[a]	%[b]	N	%[c]	N	%[c]	P-value
Geographic strata							.0013
Strata 1	27	8.2	17	56.1	10	43.9	
Strata 2	34	14.8	21	61.6	13	38.4	
Strata 3 & 4	107	77.1	37	28.5	70	71.5	
Gender							.2034
Male	61	47.6	23	29.6	38	70.4	
Female	106	52.4	52	41.4	54	58.6	
Race							.8635
White	163	98.7	72	35.0	91	65.0	
Non-White	3	1.3	1	30.2	2	69.8	
Current age							.0425
18-49	20	25.4	6	21.1	14	78.9	
50-64	50	32.3	17	26.7	33	73.3	
65+	96	42.6	50	49.3	46	50.7	
Age at cancer diagnosis							.0487
18-49	56	49.1	22	25.9	34	74.1	
50-64	74	34.6	34	38.9	40	61.1	
65+	35	16.3	17	54.4	18	45.6	
Year since diagnosis							.6313
0-5	57	45.4	22	31.1	35	68.9	
6-10	35	17.6	13	36.4	22	63.6	
10+	71	37.0	37	40.6	34	59.4	
Occupational status							.0610
Employed	26	19.9	12	38.1	14	61.9	
Retired	88	38.2	46	48.7	42	51.3	
Disabled	32	25.7	9	16.2	23	83.8	
Other	18	16.2	5	27.5	13	72.5	
Marital status							.5511
Married, living as married	99	70.6	45	34.0	54	66.0	
Divorced, separated	40	15.7	14	29.1	26	70.9	
Single, widowed	27	13.8	14	46.3	13	53.7	
Educational attainment							.0475
High school or Less	64	38.7	26	38.1	38	61.9	
Some college	53	36.3	20	23.1	33	76.9	
College graduate or more	48	25.0	27	51.4	21	48.6	
Insurance source							.2881
Private	49	38.7	25	42.0	24	58.0	
Medicare	72	40.1	30	33.6	42	66.4	
Other public	24	18.5	8	21.6	16	78.4	
None	2	2.7	0	0	2	100.0	
Household income							.0065
$0 - $19,999	43	24.9	10	14.6	33	85.4	
$20,000 - $49,999	46	33.6	21	29.0	25	71.0	
$50,000 - $99,999	41	27.5	21	40.9	20	59.1	

(Continues)

TABLE 2 (Continued)

	Total		No financial distress		Financial distress		
	N[a]	%[b]	N	%[c]	N	%[c]	P-value
$100,000+	18	14.0	12	63.3	6	36.7	
Could not see a doctor because of cost							.0065
No	150	91.5	72	38.6	78	61.4	
Yes	13	8.5	2	10.8	11	89.2	
Feelings about present household's income							<.0001
Living comfortably	55	29.8	40	69.0	15	31.0	
Getting by	51	31.2	24	36.7	27	63.3	
Finding it difficult	39	21.4	7	15.5	32	84.5	
Finding it very difficult	18	17.5	1	3.6	17	96.4	

[a]Unweighted sample size.
[b]Weighted column percentage.
[c]Weighted row percentage.

the sample (52% vs. 48%), and nearly all of the Appalachian cancer survivors were White (99%). Forty-three percent of the cancer survivors were 65 years of age or older at the time of the survey administration; correspondingly, approximately 40% of the sample, respectively, indicated they were retired and had Medicare as their primary source of health insurance. Among Appalachian Kentucky cancer survivors, 39% had a high school education or less; 9% reported cost as a barrier to seeing a physician; one-quarter reported an annual household income of ≤$19,999; and nearly 40% perceived it difficult or very difficult to get by on their present income.

In the bivariate analysis of ASK data (Table 2), geographic strata was significantly correlated with cancer-related financial distress among Appalachian Kentucky survivors with higher percentages of distress reported among residents of Strata 3 and 4 (p=.0013). There was also a significant association between financial distress and household annual income (p=.0065); as income decreased, financial distressed increased. Similarly, higher rates of cancer-related distress were observed among respondents who found it difficult or very difficult to get by on their present income (p<.0001). Individuals reporting cost as a barrier to seeing a doctor also reported higher financial distress (p=.0065). Respondents with "some college" as their highest level of education reported the highest levels of financial distress compared to college graduates and those with a high school education or less (p=.0475). Current age and age at cancer diagnosis both trended toward younger individuals (i.e., ages 18-49) reporting more financial distress. Gender, race, years since cancer diagnosis, occupational status, marital status, and insurance source were not significantly associated with cancer-related financial distress among Appalachian Kentucky survivors.

4.2 | HINTS findings

The comparison results from HINTS were noticeably different from the ASK findings (Table 3). The majority of HINTS respondents (85%) resided in Strata 1 counties. Almost 60% of the sample was female; 83% White; and 49% age 65 or older. Among HINTS cancer survivors, health insurance coverage was high at 97% and over one-quarter of the sample (27%) reported an annual household income of $100,000 or more. Among HINTS respondents, geographic strata were not significantly associated with cancer-related financial distress. Only health insurance coverage (p<.0001) and household income (p=.0035) were significantly associated with financial distress among HINTS cancer survivors.

4.3 | Comparison of ASK and HINTS findings

Table 4 showcases the differences in cancer-related financial distress across the two surveys and by geographic strata. ASK cancer survivors were significantly more likely to report financial distress (64%) compared to HINTS cancer survivors (37%, p<.0001). Specifically, there are notable differences among cancer survivors residing in Strata 3 and 4 counties: in Strata 3 and 4, 72% of ASK respondents reported cancer-related financial distress as compared to 41% of HINTS respondents (p=.0174).

Table 5 presents the multivariate analysis of cancer-related financial distress for both ASK and HINTS. For ASK, geographic strata did not retain significance in the final model (p=.355), although residing in Strata 3 and 4 indicates increased odds of financial burden as compared to residing in Strata 1 (and Strata 2). Male cancer survivors in Appalachian Kentucky were three times more likely to report financial distress compared to females (p=.015); younger and middle age respondents were almost five times more likely to report financial hardship compared to survivors age 65 and older (p=.006); and cancer survivors reporting an annual household income of ≤$19,999 were 20 times more likely to report financial distress as compared to the reference category of $100,000 or more (p=0.001). In the HINTS multivariate analysis, geographic strata was not associated with financial distress (p=.983). Although not as striking at the ASK analysis, HINTS

TABLE 3 HINTS: correlation between cancer-related financial distress and participant characteristics

	Total		No financial distress		Financial distress		
	N[a]	%[b]	N	%[c]	N	%[c]	P-value
Geographic strata							.8364
Strata 1	385	84.6	245	64.1	140	35.9	
Strata 2	22	4.8	11	60.1	11	39.9	
Strata 3 & 4	39	10.6	22	59.0	17	41.0	
Gender							.8185
Male	163	41.5	109	64.1	54	35.9	
Female	254	58.5	150	62.6	104	37.4	
Race							.2865
White	340	82.7	226	65.2	114	34.8	
Black	54	11.2	21	51.8	33	48.2	
Other	29	6.1	16	51.0	13	49.0	
Current age							.3206
18-49	31	12.5	18	50.3	13	49.7	
50-64	150	38.3	83	66.2	67	33.8	
65+	237	49.2	158	65.5	79	34.5	
Age at cancer diagnosis							.2355
18-49	159	45.7	97	62.7	62	37.3	
50-64	168	32.9	92	58.4	76	41.6	
65+	104	21.5	79	72.5	25	27.5	
Year since diagnosis							.1955
0-5	130	28.3	78	59.7	52	40.3	
6-10	83	21.0	45	55.7	38	44.3	
10+	196	50.7	130	68.9	66	31.1	
Occupational status							.0982
Employed	136	40.8	82.0	62.6	54.0	37.4	
Retired	220	44.4	147.0	67.8	73.0	32.2	
Disabled	31	6.6	11.0	36.9	20.0	63.1	
Other	32	8.2	23.0	73.1	9.0	26.9	
Marital status							.1902
Married, living as married	247	68.7	163	68.0	84	32.0	
Divorced, separated	81	13.1	43	54.2	38	45.8	
Single, widowed	98	18.2	60	60.1	38	39.9	
Educational attainment							.2573
High school or less	125	31.1	73	56.1	52	43.9	
Some college	135	37.0	81	66.1	54	33.9	
College graduate or more	171	31.9	115	67.9	56	32.1	
Health insurance coverage							<.0001
Yes	424	96.8	267	64.7	157	35.3	
No	14	3.2	4	10.6	10	89.4	
Household income							.0035
$0 - $19,999	79	14.5	35	38.4	44	61.6	
$20,000 - $49,999	118	22.4	71	61.9	47	38.1	
$50,000 - $99,999	140	36.3	96	68.6	44	31.4	
$100,000+	107	26.9	75	70.8	32	29.2	

[a]Unweighted sample size.
[b]Weighted column percentage.
[c]Weighted row percentage.

TABLE 4 Comparison of cancer-related financial distress by survey source and geographic strata

	ASK				HINTS				
	No financial distress		Financial distress		No financial distress		Financial distress		
	N[a]	%[b]	N	%	N	%	N	%	P-value
Geographic strata									
Strata 1	17	56.1	10	43.9	245	64.1	140	35.9	.9440
Strata 2	21	61.6	13	38.4	11	60.1	11	39.9	.3152
Strata 3 & 4	37	71.5	70	71.5	22	59.0	17	41.0	.0174
Total	75	35.6	93	64.4	278	63.4	168	36.6	<.0001

[a]Unweighted sample size.
[b]Weighted column percentage.

TABLE 5 Multivariate analysis identifying participant characteristics associated with cancer-related financial distress across both surveys

	ASK				HINTS			
Factors	OR	95% LCL	95% UCL	P-value	OR	95% LCL	95% UCL	P-value
Geographic strata				0.355				.983
Strata 2	0.72	0.18	2.85		1.10	0.37	3.27	
Strata 3 & 4	1.99	0.69	5.75		1.03	0.46	2.29	
Strata 1	ref				ref			
Gender				0.015				.881
Male	3.11	1.26	7.68		1.04	0.59	1.86	
Female	ref				ref			
Current age				0.006				.238
18-49	4.72	1.38	16.12		2.67	0.95	7.55	
50-64	4.78	1.63	14.07		1.22	0.68	2.19	
65+	ref				ref			
Household income				0.001				.005
$0 - $19,999	20.24	4.17	98.21		4.03	1.72	9.45	
$20,000 - $49,999	6.49	1.36	30.94		1.71	0.78	3.73	
$50,000 - $99,999	5.35	1.16	24.69		1.13	0.54	2.35	
$100,000+	ref				ref			

OR, odds ratio; LCL, lower confidence limit; UPL upper confidence limit.

respondents with an annual household income of ≤$19,999 were four times more likely to report cancer-related financial distress as compared to survivors earning $100,000 or more (p=.005).

5 | DISCUSSION

Given simultaneous national agendas focused on increasing NCI-designated cancer centers' understanding of their local communities,[29] understanding and remediating financial toxicity among cancer survivors,[35] and increasing rural cancer control research,[36,37] we explored perceptions of cancer-related financial distress among cancer survivors residing in Appalachian Kentucky with a specific focus on rurality. To our knowledge, this is among the first estimates of cancer-related financial distress representative of this unique geographic population. Moreover, the current study compared estimates from Appalachian Kentucky to nationally representative results from HINTS, illustrating [1] notable differences in the prevalence of cancer-related financial distress between the two populations and [2] underrepresentation of subpopulations of rural and extremely rural respondents in national surveys.

Our primary aim focused on estimating the prevalence and identifying predictors of cancer-related financial distress among cancer survivors in Appalachian Kentucky. Specifically, a correlation between increasing rurality as defined by geographic strata and financial distress was observed, with cancer survivors residing in rural and very rural communities experiencing more financial distress; however, this finding was not significant in the multivariate analysis after controlling for socioeconomic characteristics. The non-significant results are likely attenuated by the significant impact of high poverty and unemployment rates and low levels of educational attainment in these Appalachian Kentucky communities.[27] Similar attenuated effects were

also observed by Palmer et al. in their analysis of NHIS data comparing rural and urban differences in forgoing health care after cancer because of cost, stratified by age.[18] Interestingly, Strata 1, which is considered metro in classification, and Strata 2, -representing more urban/suburban counties, were similar in findings for cancer-related financial distress (i.e., 44% and 39%) as well as other ASK survey items (data not shown); the same pattern was observed for Strata 3 and 4 (i.e., 72%). This finding is interesting given the common grouping of RUCC codes 1-3 as metro and 4-9 as non-metro[30] and points to the need to explore more granular rural categorizations.

Similar to other studies,[3,4,18,38] younger and middle age Appalachian cancer survivors reported experiencing cancer-related financial distress more so than older adults; this was true for both current age and age at diagnosis in the bivariate analysis and current age in the final model. We posit that older individuals may be more likely to have stable health insurance coverage through Medicare; may be retired and therefore not at-risk for job loss and salary reductions; may not have the same family responsibilities as a younger individual; and/or may have amassed more assets over their lifetime, which help to offset a financial crisis following a cancer diagnosis.[4] More research is needed to fully understand financial hardship among working-age cancer survivors, particularly those in rural communities, given the significant implications of steady employment, monetary earnings, and consistent insurance coverage.[4,38]

Within the ASK data, an inverse relationship between three financially-related variables – annual household income, perceptions of household financial security, and cost as a barrier to health care – was found with cancer-related financial distress, which is commonly reported in the literature.[3,4,18] Interestingly, Appalachian cancer survivors with "some college" education were more likely to experience financial distress. Related, in New Mexico, McDougall et al. reported that colorectal cancer survivors with "some college" were almost three times more likely to be non-adherent to surveillance colonoscopy as compared to those with a college degree.[5] We posit that this group of survivors may represent the working poor and/or the under-insured, whereby they may be employed in jobs that do not require a college education, pay lower wages, and/or do not offer employer-sponsored benefits such as paid-time off and health insurance. These individuals may not have the financial security of those with higher socioeconomic or education levels, while at the same time, they may not qualify for governmental safety-net programs. This unique finding – which was not observed in the HINTS data – warrants further investigation.

In comparing ASK and HINTS findings, the overall prevalence of cancer-related financial distress was notably lower among national respondents as compared to Appalachian Kentucky estimates and the only statistically significant variable common to both datasets was the impact of annual household income on cancer-related financial distress. Interestingly, among HINTS respondents, financial distress percentages across the geographic strata were similar, ranging from 36% in Strata 1 to 41% in Strata 3-4, which is in contrast to the ASK findings discussed earlier. Clearly, ASK and HINTS focused on very different populations. For example, 8% of the ASK sample resided in Strata 1 counties compared to 85% of the HINTS sample. Cancer survivors residing in rural and completely rural counties were largely underrepresented in the HINTS dataset (i.e., only 39 respondents or approximately 10% of the weighted sample resided in Strata 3 and 4), whereas over three-quarters of the ASK sample resided in these smaller geographic areas. Selection bias, or the background populations represented by the two surveys, likely contributed to the differences in findings between the two populations, even after controlling for other variables. Underrepresentation of rural populations and exclusion of smaller counties in national surveys has been previously reported, including discussion of the potential bias introduced into national estimates and the negative implications for cancer disparities-related research and policy.[39,40]

5.1 | Strengths and limitations

There are many strengths to the current study including a focus on unique geographic population of Appalachian Kentucky, which has not been previously reported; the inclusion of extremely rural counties with a large sample size in the ASK dataset; the use of the same questions and similar methodologies across both surveys; analysis of two population-based surveys reflective of both Appalachian Kentucky and the U.S.; and the use of perceived financial distress as a proxy measure for one domain of quality of life. However, there are noted limitations that should be addressed. First, both ASK and HINTS are cross-sectional surveys, thereby limiting the ability to infer causality. The ASK response rate of 25% is less than optimal, however, it is comparable to HINTS' response rate among Appalachian respondents.[32] In addition, the two surveys captured survivors' perceptions of cancer-related financial distress at one point in time with a single question; this is in comparison to the alternative of a longitudinal cohort of cancer survivors using varied time points and multiple measures of objective and subjective financial distress that would be more comprehensive in assessing the material and psychological hardships faced by cancer patients. Common issues associated with survey data may have also affected the study results, including selection bias, non-response bias, and recall errors. In addition, we are not able to ascertain exactly how and to what magnitude cancer and its treatment affected survivors' financial situation (e.g., job loss, bankruptcy) and for how long, and whether the financial hardships influenced their treatment choices, symptom burden, and/or cancer outcomes. Lastly, our work focuses on only one state within the larger 13-state Appalachian region, thereby limiting generalizability to other Appalachian areas.

6 | CONCLUSIONS

Among our sample of Appalachian Kentucky cancer survivors, almost two-thirds indicated some level of cancer-related financial distress, which is notably higher than estimates provided by HINTS[20] and other national health surveys and likely driven by the socioeconomic characteristics of extremely rural Appalachian communities. These findings

warrant further research, including qualitative assessments with Appalachian cancer survivors, family members, and oncology providers and support staff. Cancer survivors' age and economic circumstances were also important in predicting financial distress. There is an obvious need for increased financial risk and cost of care conversations between cancer patients and their health care team[41,42]; ideally, these conversations would promote shared decision-making and improved cost-related health literacy among patients.[43-45] In addition, support services such as financial and insurance counseling, patient navigation, medication assistance programs, social services, and psychosocial oncology resources are needed to help patients manage medical and non-medical financial issues related to their cancer care.[5,34,46] These services and resources are needed within both clinical and community settings, but could also be made available through connected health programs, particularly for isolated rural and Appalachian communities.[35] There is also need to understand the implications of changing health care policies on financial distress among cancer patients, particularly in a state such as Kentucky where Medicaid expansion remains tenuous.[47]

Given the cancer and socioeconomic disparities present in rural and Appalachian communities *and* the significance of financial distress on cancer outcomes and patient wellbeing, this paper serves as a call to action to further understand and intervene on these important public health challenges.

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CONFLICT OF INTEREST

The authors have stated explicitly that there are no conflicts of interest in connection with this article.

AUTHORS' CONTRIBUTIONS

All authors had full access to the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. *Conceptualization*, R.C.V., M.F.J., B.H.; *Methodology*, R.C.V., B.H.; *Software*, Q.C., F.L., B.H.; *Investigation*, Q.C., F.L., B.H., M.F.J., L.R.S.; *Formal Analysis*, Q.C., F.L., B.H.; *Data Curation*, Q.C., F.L., B.H., R.C.V.; *Writing – Original Draft*, R.C.V., M.F.J., L.R.S., B.H.; *Writing – Review & Editing*, R.C.V., L.R.S., B.H.; *Funding Acquisition*, R.C.V., B.H.

DATA SHARING AND DATA ACCESSIBILITY STATEMENT

The NCI HINTS data and methodology reports that support the findings of this study are publically available at https://hints.cancer.gov/. The UK Markey Cancer Center ASK data and methodology that support the findings of this study are available from the corresponding author upon reasonable request.

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REFERENCES

1. National Cancer Institute. Financial toxicity and cancer treatment (PDQ) - health professional version [Internet]. *National Cancer Institute: National Institutes of Health*; 2019 [updated 2019 April 15; cited 2019 July 9]. Available from: https://www.cancer.gov/about-cancer/managing-care/track-care-costs/financial-toxicity-hp-pdq.
2. Altice CK, Banegas MP, Tucker-Seeley RD, Yabroff KR. Financial hardships experienced by cancer survivors: a systematic review. *J Natl Cancer Inst*. 2017;109(2).
3. Kent EE, Forsythe LP, Yabroff KR, et al. Are survivors who report cancer-related financial problems more likely to forgo or delay medical care? *Cancer*. 2013;119(20):3710-3717.
4. Yabroff KR, Dowling EC, Guy GP Jr, et al. Financial hardship associated with cancer in the United States: findings from a population-based sample of adult cancer survivors. *J Clin Oncol*. 2016;34(3):259-267.
5. McDougall JA, Banegas MP, Wiggins CL, Chiu VK, Rajput A, Kinney AY. Rural disparities in treatment-related financial hardship and adherence to surveillance colonoscopy in diverse colorectal cancer survivors. *Cancer Epidemiol Biomarkers Prev*. 2018;27(11):1275-1282.
6. Nipp RD, Zullig LL, Samsa G, et al. Identifying cancer patients who alter care or lifestyle due to treatment-related financial distress. *Psychooncology*. 2016;25(6):719-725.
7. Ekwueme DU, Yabroff KR, Guy GP Jr, et al. Medical costs and productivity losses of cancer survivors--United States, 2008-2011. *MMWR Morb Mortal Wkly Rep*. 2014;63(23):505-510.
8. Guy GP Jr, Ekwueme DU, Yabroff KR, et al. Economic burden of cancer survivorship among adults in the United States. *J Clin Oncol*. 2013;31(30):3749-3757.
9. Guy GP Jr, Yabroff KR, Ekwueme DU, et al. Healthcare expenditure burden among non-elderly cancer survivors, 2008-2012. *Am J Prev Med*. 2015;49(6 Suppl 5):S489-S497.
10. Meneses K, Azuero A, Hassey L, McNees P, Pisu M. Does economic burden influence quality of life in breast cancer survivors? *Gynecol Oncol*. 2012;124(3):437-443.
11. Ramsey SD, Bansal A, Fedorenko CR, et al. Financial insolvency as a risk factor for early mortality among patients with cancer. *J Clin Oncol*. 2016;34(9):980-986.
12. Chan RJ, Gordon L, Zafar SY, Miaskowski C. Financial toxicity and symptom burden: what is the big deal? *Support Care Cancer*. 2018;26(5):1357-1359.
13. Lathan CS, Cronin A, Tucker-Seeley R, Zafar SY, Ayanian JZ, Schrag D. Association of financial strain with symptom burden and quality of life for patients with lung or colorectal cancer. *J Clin Oncol*. 2016;34(15):1732-1740.
14. Henley SJ, Anderson RN, Thomas CC, Massetti GM, Peaker B, Richardson LC. Invasive Cancer Incidence, 2004-2013, and Deaths, 2006-2015, in Nonmetropolitan and Metropolitan Counties - United States. *MMWR Surveill Summ*. 2017;66(14):1-13.
15. Charlton M, Schlichting J, Chioreso C, Ward M, Vikas P. Challenges of rural cancer care in the United States. *Oncology (Williston Park)*. 2015;29(9):633-640.
16. Weaver KE, Geiger AM, Lu L, Case LD. Rural-urban disparities in health status among US cancer survivors. *Cancer*. 2013;119(5):1050-1057.

17. Zahnd WE, James AS, Jenkins WD, et al. Rural-urban differences in cancer incidence and trends in the United States. *Cancer Epidemiol Biomarkers Prev*. 2018;27(11):1265-1274.
18. Palmer NR, Geiger AM, Lu L, Case LD, Weaver KE. Impact of rural residence on forgoing healthcare after cancer because of cost. *Cancer Epidemiol Biomarkers Prev*. 2013;22(10):1668-1676.
19. McNulty JA, Nail L. Cancer survivorship in rural and urban adults: A descriptive and mixed methods study. *J Rural Health*. 2015;31(3): 282-291.
20. Zahnd WE, Davis MM, Rotter JS, et al. Rural-urban differences in financial burden among cancer survivors: an analysis of a nationally representative survey. *Support Care Cancer*. 2019;27(12):4779-4786. https://doi.org/10.1007/s00520-019-04742-z
21. Pisu M, Azuero A, Benz R, McNees P, Meneses K. Out-of-pocket costs and burden among rural breast cancer survivors. *Cancer Med*. 2017;6(3):572-581.
22. Palomino H, Peacher D, Ko E, Woodruff SI, Watson M. Barriers and challenges of cancer patients and their experience with patient navigators in the rural US/Mexico border region. *J Cancer Educ*. 2017; 32(1):112-118.
23. Wenzel J, Jones RA, Klimmek R, et al. Cancer support and resource needs among African American older adults. *Clin J Oncol Nurs*. 2012; 16(4):372-377.
24. Yao N, Alcala HE, Anderson R, Balkrishnan R. Cancer disparities in rural Appalachia: incidence, early detection, and survivorship. *J Rural Health*. 2017;33(4):375-381.
25. Wilson RJ, Ryerson AB, Singh SD, King JB. Cancer incidence in Appalachia, 2004-2011. *Cancer Epidemiol Biomarkers Prev*. 2016;25(2): 250-258.
26. U.S. Cancer Statistics Working Group. U.S. cancer statistics data visualizations tool, based on November 2018 submission data (1999-2016) [Internet]. U.S. Cancer Statistics Working Group U.S. Cancer Statistics Working Group: U.S. Department of Health and Human Services, Centers for Disease Control and Prevention, and National Cancer Institute; 2019 [updated in June 2019; cited 2019 July 9]. Available from: https://gis.cdc.gov/Cancer/USCS/DataViz.html.
27. Appalachian Regional Commission. County economic status and number of distressed areas in Appalachian Kentucky, fiscal year 2018 [Internet]. *Appalachian Regional Commission*, 1-4. 2018 [cited 2019 July 9]. Available from:. https://www.arc.gov/images/appregion/economic_statusFY2018/CountyEconomicStatusandDistressAreasFY2018Kentucky.pdf
28. Appalachian Regional Commission. Creating a culture of health in Appalachia: disparities and bright spots: key findings Appalachian Kentucky [Internet]. *Appalachian Regional Commission*; 2018; 1-5[cited 2019 July 9]. Available from: https://www.arc.gov/images/appregion/fact_sheets/HealthDisparities2017/KYHealthDisparitiesKeyFindings8-17.pdf.
29. National Cancer Institute. Population health assessment in cancer center catchment areas [Internet]. *Division of Cancer Control & Population Sciences: National Cancer Institute*; 2016 [updated 2019 March 19; cited 2019 July 9]. Available from: https://cancercontrol.cancer.gov/brp/hcirb/catchment-areas.html.
30. United States Department of Agriculture Economic Research Service. Rural-urban continuum codes: documentation [Internet]. *United States Department of Agriculture Economic Research Service*; 2013 [updated 2019 April 9; cited 2019 July 9]. Available from: https://www.ers.usda.gov/data-products/rural-urban-continuum-codes/documentation/.
31. United States Census Bureau. American community survey [Internet]. *United States Census Bureau*; [cited 2019 July 9]. Available from: https://www.census.gov/programs-surveys/acs/.
32. Westat. Health information national trends survey 4 (HINTS 4), cycle 4 methodology report [Internet]. *Westat*; 2015 [cited 2019 July 9]. 1-68 Available from: https://hints.cancer.gov/docs/HINTS_4_Cycle_4_Methodology_Report.pdf.
33. Westat. Health information national trends survey 5 (HINTS 5), cycle 1 methodology report [Internet]. *Westat*; 2017 [cited 2019 July 9]. 1-51 Available from: https://hints.cancer.gov/docs/methodologyreports/HINTS5_Cycle_1_Methodology_Rpt.pdf.
34. Spencer JC, Samuel CA, Rosenstein DL, et al. Oncology navigators' perceptions of cancer-related financial burden and financial assistance resources. *Support Care Cancer*. 2018;26(4):1315-1321.
35. Federal Communications Commission. FCC-NCI broadband cancer collaboration [Internet]. *Federal Communications Comission*; 2017 [updated 2018 October 18; cited 2019 July 9]. Available from: https://www.fcc.gov/health/cancer.
36. Kennedy AE, Vanderpool RC, Croyle RT, Srinivasan S. An overview of the national cancer institute's initiatives to accelerate rural cancer control research. *Cancer Epidemiol Biomarkers Prev*. 2018;27(11): 1240-1244.
37. Blake KD, Moss JL, Gaysynsky A, Srinivasan S, Croyle RT. Making the case for investment in rural cancer control: an analysis of rural cancer incidence, mortality, and funding trends. *Cancer Epidemiol Biomarkers Prev*. 2017;26(7):992-997.
38. Banegas MP, Guy GP Jr, de Moor JS, et al. For working-age cancer survivors, medical debt and bankruptcy create financial hardships. *Health Aff (Millwood)*. 2016;35(1):54-61.
39. Bennett KJ. Rural population estimates: an analysis of a large secondary data set. *J Rural Health*. 2013;29(3):233-238.
40. Srinivasan S, Moser RP, Willis G, et al. Small is essential: importance of subpopulation research in cancer control. *Am J Public Health*. 2015; 105(Suppl 3):S371-S373.
41. Altomare I, Irwin BB, Zafar Y, Houck K, Maloney B. Physician experience and attitudes toward addressing the cost of cancer care. *J Oncol Pract*. 2016;12(3):e281-e288. 247-288
42. Irwin B, Kimmick G, Altomare I, et al. Patient experience and attitudes toward addressing the cost of breast cancer care. *Oncologist*. 2014; 19(11):1135-1140.
43. Tucker-Seeley RD, Yabroff KR. Minimizing the "financial toxicity" associated with cancer care: advancing the research agenda. *J Natl Cancer Inst*. 2016;108(5):djv410. https://doi.org/10.1093/jnci/djv410
44. Meropol NJ, Schrag D, Smith TJ, et al. American Society of Clinical Oncology guidance statement: the cost of cancer care. *J Clin Oncol*. 2009;27(23):3868-3874.
45. Zafar SY. Financial toxicity of cancer care: it's time to intervene. *J Natl Cancer Inst*. 2016;108(5):djv370. https://doi.org/10.1093/jnci/djv370
46. Rutledge TL, Kano M, Guest D, Sussman A, Kinney AY. Optimizing endometrial cancer follow-up and survivorship care for rural and other underserved women: patient and provider perspectives. *Gynecol Oncol*. 2017;145(2):334-339.
47. Campo-Flores A. Before repeal and replace, Kentucky is dismantling Health Law [Internet]. *Wall Street Journal*. 2018 [cited 2019 July 9]. Available from:. https://www.wsj.com/articles/before-repeal-and-replace-kentucky-is-dismantling-health-law-1497178801

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of this article.

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